A4.3.5 Unsupervised learning: Association

4.3.5 Association

A4.3.5 explains association rule learning, used to uncover relations between different attributes in large datasets. For instance, in crime analysis, it can reveal links like high vandalism rates often co-occurring with theft, aiding predictive policing and resource allocation

Introduction

Continuing our journey into Unsupervised Learning, we move from finding *groups* of similar items (Clustering) to finding *relationships* between different items. **Association Rule Mining** is a powerful technique used to discover interesting and useful relations between variables in large datasets.

What is Association Rule Mining?

Association Rule Mining is a rule-based machine learning method for discovering interesting relations between variables in large databases. It is intended to identify strong rules discovered in databases using some measures of interestingness.

- **The Goal:** To find "if-then" rules that describe how often items appear together in a transaction.
- **Analogy:** It's like being a detective for data. You sift through millions of receipts from a supermarket to find clues about what people buy together.

The most famous (though possibly mythical) example is the discovery that "customers who buy diapers are also very likely to buy beer." This kind of insight can be incredibly valuable.

The Anatomy of an Association Rule

An association rule has two parts: an **antecedent** (the "if" part) and a **consequent** (the "then" part).

{Antecedent} -> {Consequent}

- **Antecedent:** An item or set of items found in the data.
- **Consequent:** An item or set of items found in combination with the antecedent.

Example: `{Bread, Butter} -> {Milk}`

Interpretation: "Customers who buy bread AND butter are also likely to buy milk."

Interpreting the Results: Is a Rule Interesting?

Not all rules are useful. The rule `{Bread} -> {Milk}` is not very surprising. To find truly interesting rules, data scientists use several measures, including:

- **Support:** How frequently the items in the rule appear together in the dataset. (e.g., "What percentage of all transactions contain both bread and milk?"). A high support means the rule applies to a large number of cases.
- **Confidence:** The probability that the consequent will be purchased given that the antecedent has been purchased. (e.g., "Of all the customers who bought bread, what percentage *also* bought milk?"). A high confidence means the rule is very reliable.

The goal is to find rules with both high support and high confidence.

Real-World Applications

Market Basket Analysis

This is the classic application, used by retailers like Amazon and Walmart to understand customer purchasing behaviour.

- **Store Layout:** Placing items that are frequently bought together (like chips and salsa) near each other to increase sales.
- **Product Bundling:** Creating special offers or bundles for items that are often purchased together (e.g., a "movie night" bundle with popcorn, soda, and candy).
- **Online Recommendations:** Powering the "Frequently Bought Together" and "Customers who bought this also bought..." features on e-commerce websites.

Predictive Policing

Association rule mining can be used in law enforcement to discover patterns in crime data.

- **Scenario:** A police department analyses crime reports from different city areas.
- **Association Rule in Action:** The mining technique might reveal a rule like: `{Reports of Vandalism, Streetlight Outages} -> {Incidents of Theft}`.
- **Interpretation:** This rule suggests that areas with a high frequency of both vandalism and broken streetlights are also highly likely to experience thefts.
- **Resource Allocation:** This insight helps law enforcement to be proactive. Instead of just reacting to thefts, they can allocate resources to fix streetlights and increase patrols in areas where the antecedent conditions are met, potentially preventing thefts before they happen.

Flashcards

Use this Quizlet to learn all the key terms in this section.

Match

A 4.3.5 Unsupervised learning: Associati...

Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

View this flashcard set

Choose a Study Mode

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

Association Rule Mining is an example of which main type of machine learning?

A. Reinforcement Learning

B. Transfer Learning

C. Unsupervised Learning

D. Supervised Learning

2

What is the primary goal of Association Rule Mining?

A. To classify data based on pre-existing labels.

B. To discover "if-then" rules that describe relationships between items in a dataset.

C. To group similar data points into clusters.

D. To predict a continuous numerical value.

3

In the association rule {A} -> {B}, 'A' is known as the:

A. Antecedent

B. Centroid

C. Consequent

D. Support

4

In the association rule {Coffee} -> {Sugar}, the "confidence" of the rule measures:

A. The percentage of all transactions that contain both coffee and sugar.

B. The total number of times coffee was sold.

C. The popularity of coffee.

D. The percentage of customers who bought coffee that also bought sugar.

5

The classic real-world application of using association rules to analyze customer transaction data is called:

A. Collaborative Filtering

B. Predictive Policing

C. Customer Segmentation

D. Market Basket Analysis

6

A rule with high "support" means that:

A. The rule was discovered by a supervised algorithm.

B. The antecedent and consequent are not related.

C. The rule is very reliable and almost always true.

D. The items in the rule appear together frequently in the dataset.

7

Which of the following is the best interpretation of the rule {Rainy Day} -> {Umbrella Sale}?

A. Selling umbrellas causes it to rain.

B. All umbrella sales happen on rainy days.

C. When the day is rainy, it is also likely that an umbrella sale will occur

D. On most days, umbrellas are sold.

8

Predictive policing uses association rules to:

A. Identify patterns and environmental factors that are correlated with crime to help alloc

B. Predict the exact time and location of the next crime.

C. Cluster criminals into different groups.

D. Determine if a suspect is guilty or innocent.

9

In the rule {A, B} -> {C}, the antecedent is:

A. {A}

B. {A, B}

C. There is no antecedent.

D. {C}

10

A data scientist finds the rule {Hot Dogs} -> {Buns}. This rule likely has:

A. Low support and high confidence.

B. High support and high confidence.

C. Low support and low confidence.

D. High support and low confidence.

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